Basic Principles of Medicine 1
Module : CPR Module
DLA

Applied ECG: Interpretation of abnormal ECG

Video Link to this DLA:

https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=acceb3e4-7ee2-4822-ae7e-b0fa01140e54

Read this DLA after CPR Lectures: Basis of ECG I and II

Dr. Subramanya Upadhya
(Email: supadhya@sgu.edu)

Copyright

Learning Objectives: DLA on Applied ECG

SOM.MK.II.BPM1.3.CPR.2.PHYS.0178. Describe the alteration in conduction responsible for most common arrhythmias: tachycardia, bradycardia, A-V block, atrial and ventricular fibrillation.

Recommended Reading

1. Rhoades RA, Bell DR. Medical Physiology. Principles for Clinical Medicine. Sixth Edition. Wolters Kluwer, 2023.

2. Goldberger's Clinical Electrocardiography, A simplified approach. Ninth Edition. Goldberger, Ary L., MD, FACC. 2018 Elsevier Inc.

ECG INTERPRETATION (How to read an ECG: steps to follow)

Follow the following 8 steps:

Step 1. Rate: Find the heart rate – Is in normal range? (Normal range is between 60 and 100 bpm)

Step 2. Rhythm: Is the rhythm regular or irregular?

Step 3. Axis: Is there left or right axis deviation?

Step 4. Intervals: What is the duration of the PR interval? What is the QRS interval? What is the QT interval?

Step 5. P wave: What is the height, duration and polarity of the P wave?

Step 6. QRS complex: Is the QRS duration normal or wide?

Step 7. ST segment and T wave: Is there ST elevation or depression compared to the TP segment? Are the T waves inverted?

Step 8. Overall interpretation: Give **summary** of the steps, 1-7.

Normal Sinus Rhythm

Lead II rhythm strip: Sinus rhythm

An ECG of a normal healthy person is said to be a sinus rhythm. The features of a sinus rhythm in ECG are:

- Heat Rate: 60 to100/min.
- Heart rhythm: Regular R-R interval.
- QRS axis (Mean electrical axis): Range between -30 to +90 degrees.
- Intervals: PR int= 3-5 small boxes; QRS int= 2-3 small boxes and QT int= 8-12 small boxes.
- P waves: Positive in all leads except aVR and all QRS complexes are preceded by a P wave (P:QRS ratio is 1:1)
- Segments: ST-segment and TP-segments are on the base line (isoelectric line).
- T wave: Positive in all leads except aVR.

Sinus rhythm Indicates: Electrical signal is generated by SA node and cardiac impulse is conducted via normal conduction pathways with normal AV-delay.

Definition and Types of Cardiac Arrhythmias

Definition: Arrhythmias are electrical abnormalities of the heart, either due to an abnormal rate or rhythm or both.

List of most common arrhythmias:

- Bradyarrhythmia: Heart rate is less than 60/min. Example: Sinus bradycardia.
- Tachyarrhythmia: Heart rate is more than 100/min Example: Sinus tachycardia.
- Conduction blocks (Heart blocks): Defective conduction of impulse from atria to ventricles through AV node. Examples: First-degree, second-degree & third-degree heart blocks.
- Atrial fibrillation: Multiple foci of impulse generation in the atria.
- Ventricular fibrillation: Multiple foci of impulse generation in the ventricles.

ECG in Sinus Bradycardia

- Everything is NORMAL in ECG except the **heart rate is less than 60/min**.

- What is the heart rate in the above ECG?
- Heart rate = 300/5.8 large boxes = 52/min.

SA node (pacemaker) generates impulse for HR, but at a **slower pace** than normal. This is due to slow rate of depolarization of phase 4 of SA node action potential.

- Sinus bradycardia occurs in vagus nerve stimulation, carotid massage, hypothyroidism, and patients who take beta blockers.

ECG in Sinus Tachycardia

- Everything is NORMAL in ECG except the **heart rate is more than 100/min**.

- What is the heart rate in the above ECG?
- Heart rate = 300/2.4 large boxes = 125/min.

SA node (pacemaker) generates impulse for HR, but at a **faster pace** than normal. This is due to increased rate of depolarization of phase 4 of SA node action potential.

- Sinus tachycardia occurs in sympathetic stimulation, during exercise, hyperthyroidism, fever, and patients who take beta agonists.

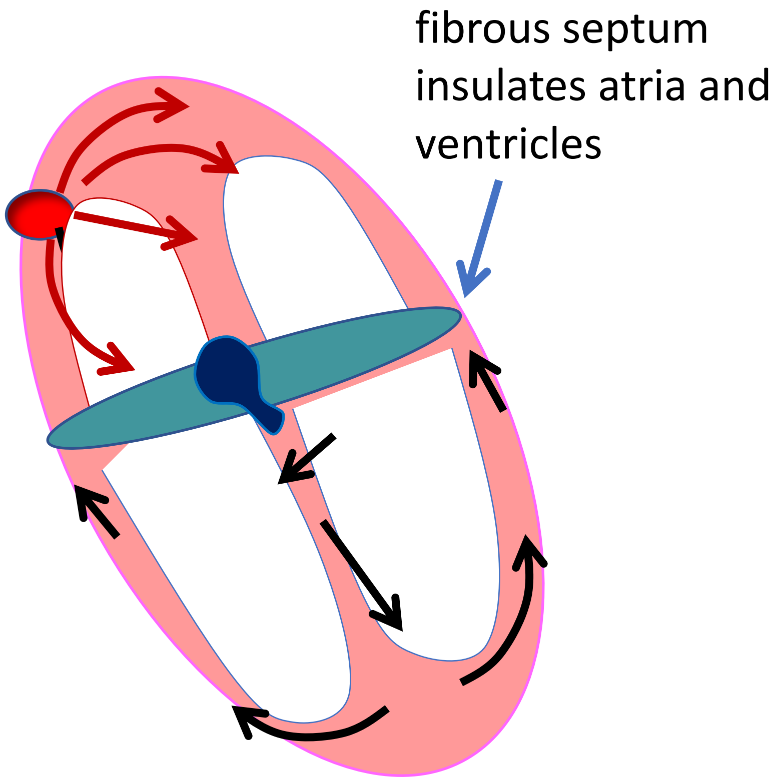
Conduction blocks (also known as Heart Blocks)

Conduction blocks occur mostly in the AV node: Why

•The ventricles are electrically separated from the atria by a fibrous septum
•The only electrical pathway to pass the impulses from the atria to the ventricles is through the AV node
•The AV node delays the impulse by about 0.1 sec (P-R interval), giving atria time to contract and fill ventricles.

•Why does the AV node slow the signals?

1. Small cells: The wave of depolarization has to jump many cell membranes and moves slowly. Greater resistance
2. Few gap junctions between the cells.
3. High resting membrane potential (RMP, – 60 mV). Thus most voltage gated Na channels are closed. AV node depolarization is by opening of L-type Ca^{2+} channels which is a slower process.



ECG in AV conduction block (Heart block)

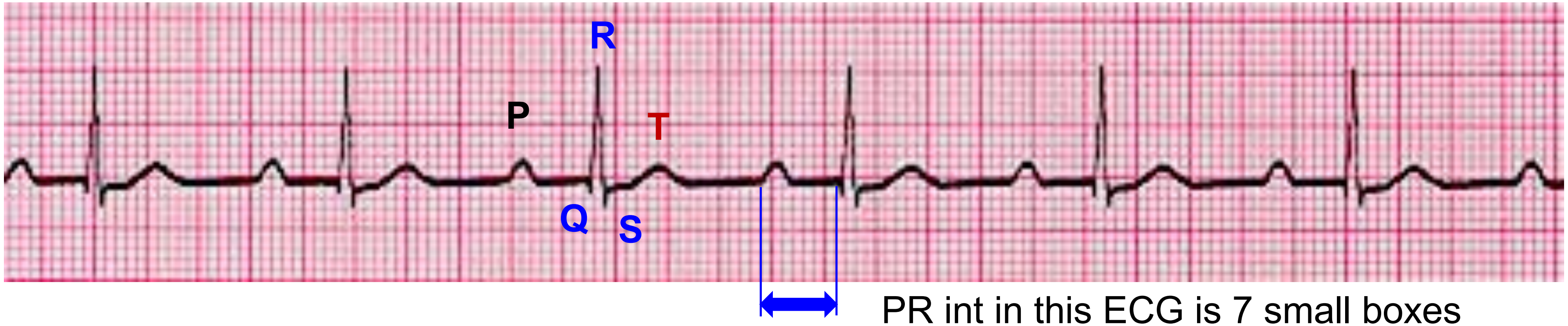
- There are 3 severities of heart blocks:
- First degree, second degree & third-degree blocks.

- First degree heart block: PR is longer than normal but all impulses arriving at AV node pass to the ventricles.

- Second degree block: PR interval progressively prolong in successive beat until some impulse fails to pass through the AV node to the ventricles.

- Third degree heart block: Absolutely NO impulses pass through the AV node to reach ventricles. This is called complete AV block.

ECG in First-degree heart block



PR int in this ECG is 7 small boxes

- Heart rate is normal or slow (In the above ECG, HR is = 300/4.8 = 62/min).
- All ECG waves are normal.
- PR interval is **more than 5 small boxes**.
- Prolonged PR interval is the only ECG finding in first-degree heart block.
- It occurs in old age, coronary artery disease & diseases that produce scar tissue.

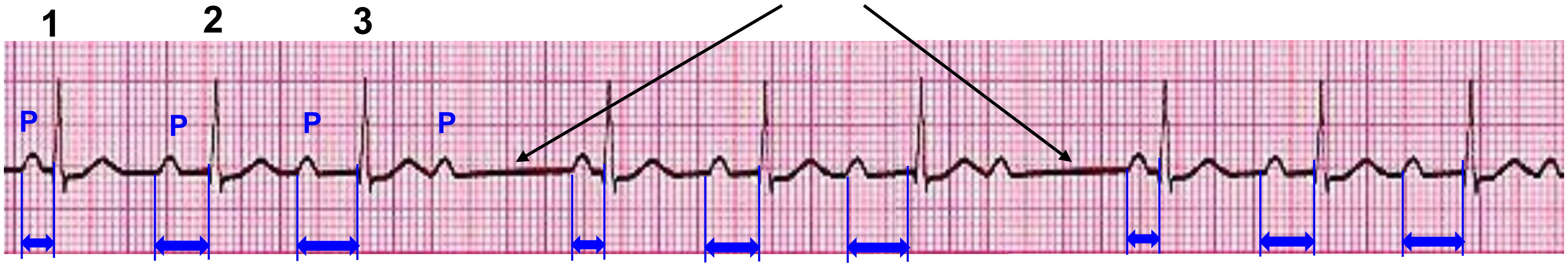
In the above ECG, PR interval = 7 small boxes
= 7x0.04
= 0.28 sec.
(Normal PR int= 0.12 to 0.20 sec)

ECG in Second-degree heart block

There are two ECG patterns: Mobitz Type I and Mobitz Type II

In both, AV node can conduct impulses normally, but fails to conduct impulses at certain time. This produces an ECG pattern where not all P waves are accompanied by a QRS complex.

2nd degree heart block: Mobitz type I (also known as Wenckebach phenomenon):
The PR interval lengthens progressively from one beat to the next and ends up in no impulse passing to ventricle (QRS absent).

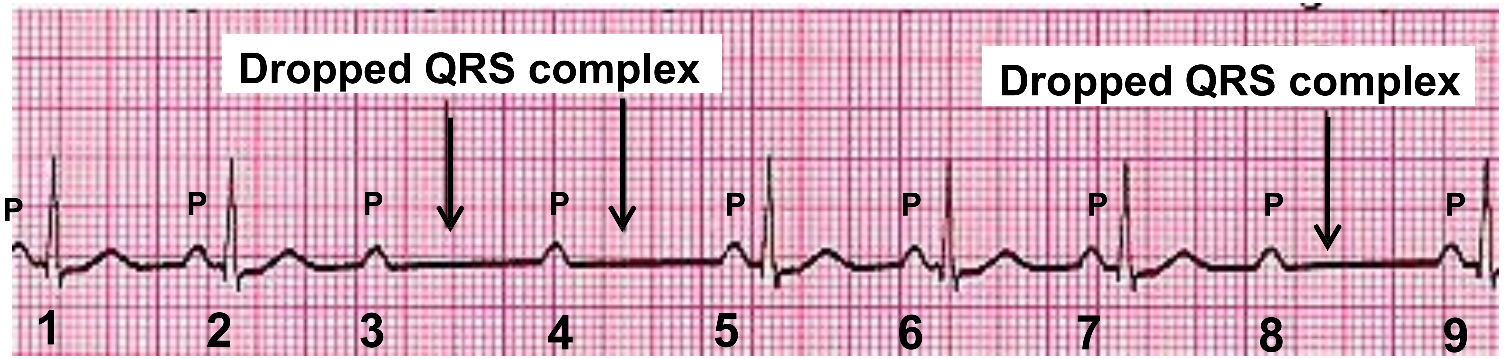


In the above ECG, note that the PR interval progressively increases in the first three beats (numbers, 1, 2 & 3) and the **fourth P wave does not have an accompanying QRS wave**. This pattern continues giving a missed beat (QRS complex) at times.

ECG in Second-degree heart block (continued from previous slide)

2nd degree heart block: Mobitz type II :

There is no progressive lengthening of the PR interval from one beat to the next. Most of the impulse pass through the AV node but there are random drops in QRS.



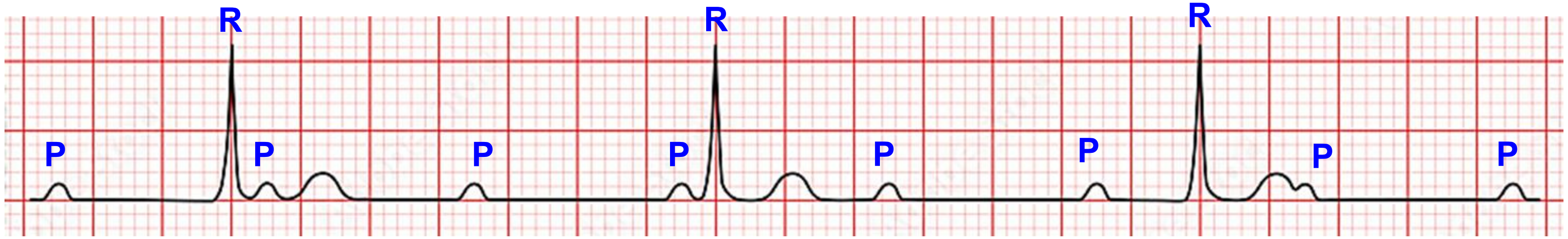
In the above ECG, note that the first two impulses pass through normally (both have P wave accompanying QRS), but the third and fourth P waves do not have a QRS. QRS drop is shown with arrows. Again, 5th, 6th and 7th impulses pass through normally, and 8th impulse does not pass through the AV node. This pattern produces an irregular heart rhythm.

ECG in Third-degree heart block

Third-degree block is also known as complete atrioventricular block.

Here, no impulses from atria reach ventricles. The AV node does not transmit any impulses (**complete block** in conduction). Ventricles generate their own impulses.

This produces an ECG pattern where no P waves are accompanied by a QRS complex. Both P and QRS are totally unrelated to each other.



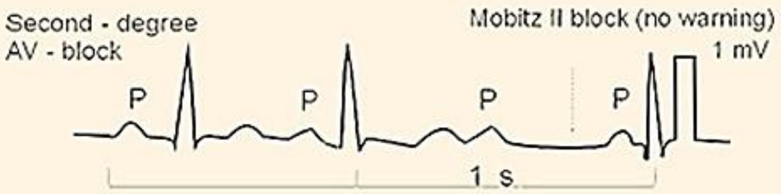
In the above ECG, note that **none of the P waves accompany QRS.** Both P and QRS occur independently. However, P waves occur at a regular interval (interval between two successive P waves is the same throughout). QRS also occur at a regular interval. Therefore, patient with complete heart block has a slow heart rate but regular rhythm (Heart rate in above ECG is 42/min). Such patient require an artificial pacemaker implanted to have a normal heart rate.

Summary of Heart Blocks (The Heart block Poem, adapted from Princeton surgical gp)

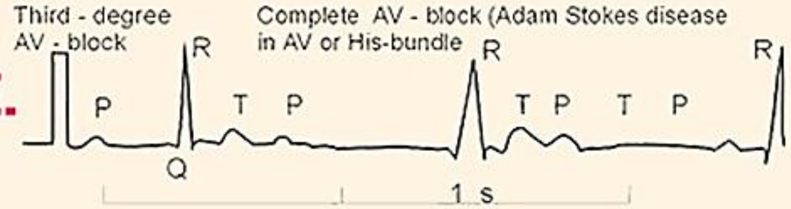
If the **R** is far from **P**,
then you have a **FIRST DEGREE**.

Longer, longer, longer, drop!
Then you have a **WENKEBACH**.

If some **P**s don't get through,
then you have **MOBITZ II.**

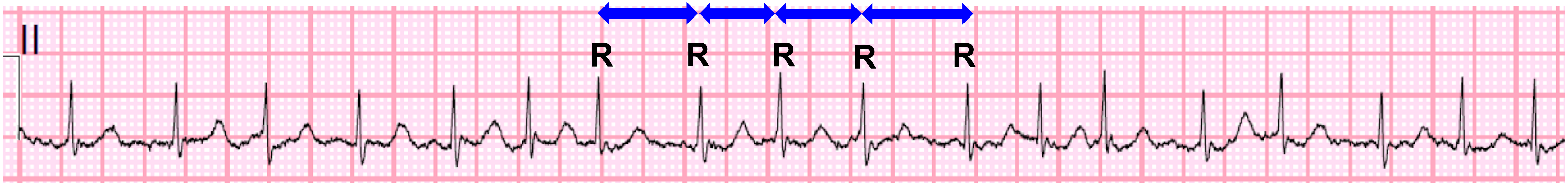


If **P**s and **Q**s don't agree,
then you have a **THIRD DEGREE.**



ECG in Atrial Fibrillation

Multiple sites generate electrical impulses in atria. There is no single wave of depolarization spreading over the atria.



ECG findings:

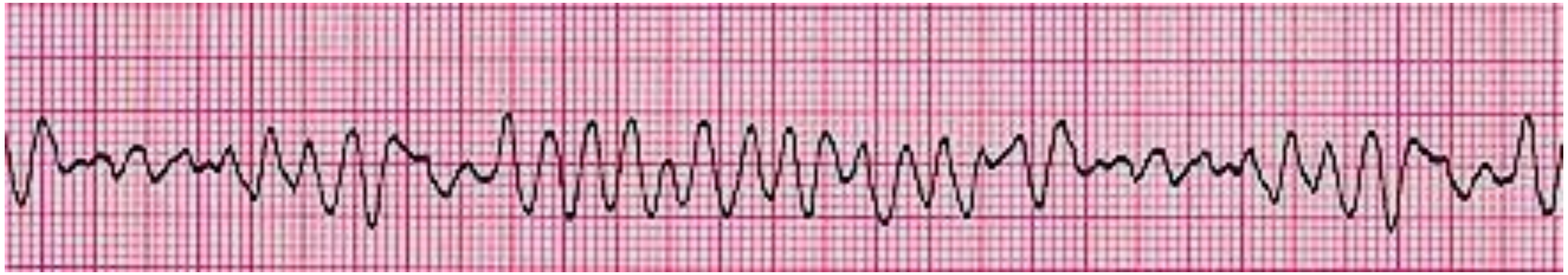
i) P waves are absent in Atrial fibrillation.

ii) **Irregular R-R intervals**. Due to refractory period of AV node only some signals pass through AV node to ventricles at irregular intervals. This produces an irregular heart rhythm.

The atria do not contract; There is no atrial systole (absent “a” wave in JVP); The last part of ventricular filling from atrial kick is absent (absent S4 sound)!

ECG in Ventricular Fibrillation (VF)

Multiple sites generate electrical impulses in the ventricles. There is no single wave of depolarization spreading over the ventricles.



Defibrillation required!

The ECG in VF shows rapid (300 to 400/min), **irregular, shapeless** QRST **undulations** of variable amplitude, morphology, and interval.

Ventricles do not contract; therefore, **NO cardiac output and NO recordable BP!** Patients invariably present with sudden collapse (syncope and/or sudden cardiac arrest).

Practice Question 1:

A 69-year-old woman comes to the physician because of a 3-week history of fatigue and light-headedness. Her blood pressure is 114/72 mm Hg. Physical examination shows regularly irregular pulse. An ECG in lead II is shown below. Which of the following ECG abnormalities is most likely in this patient?

A. PR interval is prolonged
B. P waves are absent
C. QRS interval is greater than 3 small boxes
D. There is progressive lengthening of PR interval which ends in absent QRS wave
E. None of the impulses from atria reach ventricles

Answer: D.
This is a second-degree heart block, Mobitz type I

Practice Question 2:

An 82-year-old man comes to the physician because of a 4-week history of recurrent episodes of palpitations. He says each episode lasts for about 10-15 seconds and disappear on its own. He has 20-year history of type 2 diabetes mellitus. He smokes 1 pack of cigarette daily for the last 30 years. His pulse at rest is 82/min, blood pressure is 134/86 mm Hg and respirations are 16/min. Physical examination shows no abnormalities. An ECG in lead II recorded during an episode is shown below. Which of the following electrical abnormalities is most likely in this patient's heart?

A. Not all atrial impulses reach ventricles
B. QRS complexes are wide
C. SA node generates impulses at a faster rate during an episode
D. Ventricle generates its own impulses
E. He has an irregularly irregular heart rhythm during an episode

Answer: C.
Heart rate during an episode = 150/min.

(The P waves are partially buried in T waves because of a faster heart rate)

Any Questions ?

How to reach me: Either walk into my office or email.

Subramanya Upadhya, MD.
Professor,
Depart of Physiology, Neuroscience & Behavioral Sciences, Lower Charter Hall,
School of Medicine, St. George's University.

Email: supadhya@sgu.edu